

SFM-01 Summary Document

1. Overview

- Excel is a quick and cheap way to do hypothesis testing of a trading idea. For detailed backtesting, we recommend using a programming language like Python or feature-rich tools.
- This lecture primarily focuses on hypothesis testing or backtesting of trading ideas using spreadsheet software like Microsoft Excel. It has used some of the simple yet powerful and popular indicators in the examples to demonstrate the same.

2. Moving Average

- **Moving Average (Simple Moving Average):** Moving Average is a rolling average of a price series in a specific frequency, for example, please refer below to the daily price series of RELIANCE stock.
-

Date	Price
2021-05-03	1959
2021-05-04	1917
2021-05-05	1920
2021-05-07	1932
2021-05-11	1933
2021-05-12	1913
2021-05-14	1937
2021-05-17	1960
2021-05-18	1988
2021-05-19	1997
2021-05-25	1963
2021-05-26	1970
2021-05-28	2095

The moving average of the last 5 periods on 2021-05-12 is calculated as the average of prices from 2021-05-03 to 2021-05-11.

Date	Price	Moving Average(5)
2021-05-03	1959	
2021-05-04	1917	
2021-05-05	1920	
2021-05-07	1932	
2021-05-11	1933	
2021-05-12	1917	=average(C3:C7)
2021-05-14	1937	
2021-05-17	1960	
2021-05-18	1988	
2021-05-19	1997	
2021-05-25	1963	
2021-05-26	1970	
2021-05-28	2095	

The moving average of the last 5 periods on 2021-05-17 is calculated as the average of prices from 2021-05-05 to 2021-05-14.

Date	Price	Moving Average(5)
2021-05-03	1959	
2021-05-04	1917	
2021-05-05	1920	
2021-05-07	1932	
2021-05-11	1933	
2021-05-12	1913	1932
2021-05-14	1937	
2021-05-17	1967	=average(C5:C9)
2021-05-18	1988	
2021-05-19	1997	
2021-05-25	1963	
2021-05-26	1970	
2021-05-28	2095	

3. Moving Average Crossover

- Crossovers are a common way traders use moving averages. A crossover occurs when a short moving average crosses either above the long moving average or below the long moving average.
- When the short moving average is above the long moving average, a stock is considered to be bullish.
- When the short moving average is below the long moving average, a stock is considered to be bearish.
- Moving average crossovers help you determine when a trend is about to end and reverse. It is used to capture the momentum of the instrument.
- Let us consider a trading idea described in the lecture:
 - Trading Idea:
 - Use moving averages as an indicator for this strategy.
 - Short moving average = 30 DMA (Daily Moving Average)
 - Long moving average = 100 DMA (Daily Moving Average)
 - If $30\text{ DMA} > 100\text{ DMA}$ then take a long position and square off the position when $30\text{ DMA} \leq 100\text{ DMA}$
 - If $30\text{ DMA} < 100\text{ DMA}$ then take a short position and square off the position when $30\text{ DMA} \geq 100\text{ DMA}$
 - Steps:
 - Download the OHLC (Open, High, Low, Close) data from Yahoo finance in the excel file.
 - Calculate the 30 days moving average and 100 days moving average using the steps mentioned in the previous section.
 - Create the raw trading signals
`=IF(30DMA>100DMA,"BUY",IF(30DMA<100DMA,"SELL","HOLD"))`
 - Create trading signals (to reduce the generation of false signals, we consider the previous day's SMA and LMA) in the next column by comparing the values of 30 DMA and 100 DMA.
 - `=IF(AND(30DMA_PREV<=100DMA_PREV,30DMA_CURR>100DMA_CURR),"BUY",IF(AND(30DMA_PREV>=100DMA_PREV,30DMA_CURR<100DMA_CURR),"SELL","HOLD"))`
 - Excel steps:
 - Calculate SMA (30 DMA) in column H
 - Calculate LMA (100 DMA) in column I
 - Calculate trading signal as buy/sell in column J
 - Calculate trading price in column K as per the signal in column J
 - Calculate strategy returns in column L

4. VWAP

- Volume Weighted Average Price (VWAP) is an indicator calculated using volume and price.
- Generally, Typical price is used to calculate VWAP.
- $VWAP = \frac{\sum(\text{Volume} * \text{Price})}{\sum(\text{Volume})}$
- Excel steps:
 - Calculate typical price in column H. The typical price is the average of high, low, and close prices.
 - Calculate TPVolume (Total Price Volume) in column I by multiplying the typical price with the volume of the security.
 - Calculate Cumulative TPVolume in column J by adding the current TPVolume to the previous one.
 - Calculate Cumulative Volume in column K by adding the current volume to the previous one.
 - Finally, calculate VWAP in column L by dividing the cumulative TPVolume by cumulative volume.
 - Calculate trading signal as buy/sell in column M.
 - Calculate trading price in column N as per the signal in column M.
 - Calculate returns in column O.

5. ATR-based Breakout Strategy:

1. Initiate a new buy trade if the price goes above 'x' candles high
2. Initiate a new sell trade if the price goes below 'y' candles low
3. Exit trade when:
 - a. Price goes against us by 'a' times ATR (Average True Range)
 - b. Price goes in our favour by 'b' times ATR (Average True Range)
4. Only take one position at a time. Ignore new signals if there is an ongoing trade.
5. Fixed position size of 1 'quantity'.
6. Average True Range(ATR) is calculated as shown below:

True Range (TR): $\text{MAX}(H-L, H-\text{PCC}, \text{PCC}-L)$

$$\text{ATR}(i) = \frac{\text{ATR}(i-1) * (n-1) + \text{TR}(i)}{n}$$

$\text{ATR}(1) = \text{Average of previous 'n' TRs}$

Where,

- PCC: Previous Candle Close
- H: High
- L: Low

We have historical data of the instrument as:

S No.	Date	OPEN	HIGH	LOW	LAST_PRICE
1	1/10/18 9:15	10599.85	10609.40	10598.05	10607.35
2	1/10/18 9:16	10607.35	10610.40	10601.70	10607.55
3	1/10/18 9:17	10607.55	10611.05	10598.50	10608.55
4	1/10/18 9:18	10609.05	10615.80	10606.95	10611.45
5	1/10/18 9:19	10611.50	10614.65	10603.65	10604.80
6	1/10/18 9:20	10604.90	10608.95	10601.95	10605.90
7	1/10/18 9:21	10606.15	10613.25	10597.50	10599.30
8	1/10/18 9:22	10599.85	10605.45	10594.35	10597.95
9	1/10/18 9:23	10598.40	10602.90	10589.25	10591.60
10	1/10/18 9:24	10591.85	10593.40	10586.95	10591.30
11	1/10/18 9:25	10591.15	10593.65	10584.05	10591.90
12	1/10/18 9:26	10591.60	10599.55	10586.25	10594.00

To develop the model, we require some parameters. These parameters will be used in the calculations of the strategy as follows:

1. Initiate a new buy trade if the price goes above 'x' candles high -- (Parameter 'x')
2. Initiate a new sell trade if the price goes below 'y' candles low -- (Parameter 'y')
3. Exit an existing trade (position) when we get the opposite signal, and also initiate a new position in the opposite direction.

INPUT PARAMETERS		
High	3	x
Low	3	y
ATR (n)	20	n

6. RSI (Suggested Exercise)

- RSI is one of the most popular indicators used to measure the momentum of the stock market.
- The Relative Strength Index helps you decide if a stock is overbought or oversold.
- It's a momentum-based oscillator and is a widely used technical analysis tool.
- RSI compares recent upwards movements to recent downwards movements in the closing price of a stock.

RSI Calculation

- RSI is defined as follows:
 - $RSI = 100 - 100 / (1 + RS)$
 - RS is the Relative Strength, which is the ratio of average gains and losses in the given period.

RSI Interpretation:

- RSI varies between 0 and 100.
- A stock is generally considered overbought if RSI moves above 70, or oversold if its RSI moves below 30.
- If the stock price reaches new highs, but the RSI does not rise above its previous high, then the stock price is due to fall.
- When the RSI moves above 50, the average gains outweigh the average losses; this is considered bullish.
- When the RSI falls below 50, the average losses outweigh the average gains; this is considered bearish.

RSI Indicator: Stocks, Formula, Calculation and Strategies:

- <https://blog.quantinsti.com/rsi-indicator/>.

7. Quantification and Back-testing of a Trading Idea with Multiple Entry Condition Taken from Quantified Strategies:

<https://www.quantifiedstrategies.com/> is a website where one can get an initial ideation for trading strategies.

The faculty demonstrates how these strategies can be formulated and backtested on MS Excel, by using [this](#) strategy. The link now requires a paid subscription. QuantInsti has no commercial affiliation or financial interest in the platform.

Brief note:

This article demonstrates a simple day-trading setup in the XLP consumer stock.

The strategy is to buy at (or near) the day's close if

- the close-to-close returns are below a certain threshold and
- the day opened with a gap down below a certain threshold.

Underlying premise: Even defensive sectors can offer short-term mean-reversion opportunities.

8. Excel Terminologies and Functions:

- **Cell Range:** A cell range is a combination of cells with one or more rows and columns. The range can be determined by using a colon (:) between cell references or by manually selecting the rows and columns. Refer to the examples below:

Date	Open	
2021-03-29	12950.5	
2021-03-30	12904.0	
2021-03-31	12970.3	
2021-04-01	13268.9	
2021-04-05	13434.0	
2021-04-06	13567.2	
2021-04-07	13553.9	
2021-04-08	13730.7	
2021-04-09	13709.7	
2021-04-12	13792.7	
2021-04-13	13874.6	
2021-04-14	13988.9	
2021-04-15	13934.9	
2021-04-16	14036.0	
2021-04-19	13970.8	
2021-04-20	13892.0	
2021-04-21	13768.9	
2021-04-22	13920.9	
2021-04-23	13794.5	
2021-04-26	13954.9	
2021-04-27	14048.1	
2021-04-28	13953.6	
2021-04-29	14063.9	
2021-04-30	13857.8	
	=B2:B25	

The above example shows a cell range with a single column and multiple rows.

Date	Open	High
2021-03-29	12950.5	13013.5
2021-03-30	12904.0	12929.5
2021-03-31	12970.3	13167.1
2021-04-01	13268.9	13333.4
2021-04-05	13434.0	13623.6
2021-04-06	13567.2	13660.3
2021-04-07	13553.9	13648.8
2021-04-08	13730.7	13763.2
2021-04-09	13709.7	13849.3
2021-04-12	13792.7	13839.4
2021-04-13	13874.6	14002.5
2021-04-14	13988.9	13990.5
2021-04-15	13934.9	14040.1
2021-04-16	14036.0	14050.4
2021-04-19	13970.8	14025.6
2021-04-20	13892.0	13934.9
2021-04-21	13768.9	13939.7
2021-04-22	13920.9	13954.0
2021-04-23	13794.5	13989.4
2021-04-26	13954.9	14039.4
2021-04-27	14048.1	14048.3
2021-04-28	13953.6	13992.1
2021-04-29	14063.9	14073.5
2021-04-30	13857.8	13969.2
	=b2:c22	

Here, we see an example of multiple columns and multiple rows cell range.

Date	Open
2021-03-29	12950.5
2021-03-30	12904.0
2021-03-31	12970.3
2021-04-01	13268.9
2021-04-05	13434.0
2021-04-06	13567.2
2021-04-07	13553.9
2021-04-08	13730.7
2021-04-09	13709.7
2021-04-12	13792.7
2021-04-13	13874.6
2021-04-14	13988.9
2021-04-15	13934.9
2021-04-16	14036.0
2021-04-19	13970.8
2021-04-20	13892.0
2021-04-21	13768.9
2021-04-22	13920.9
2021-04-23	13794.5
2021-04-26	13954.9
2021-04-27	14048.1
2021-04-28	13953.6
2021-04-29	14063.9
2021-04-30	13857.8
	328941.8
	=SUM(B2:B25)

The above snapshot illustrates how to use a cell range in a function.

Functions:

“AND”: It is a function under the logical category. If all the conditions satisfy to be TRUE, it returns TRUE else FALSE.

Syntax:

=AND(Cond1,Cond2) returns *TRUE* or *FALSE*

“OR”: It is a function under the logical category. If either of the conditions satisfies to be TRUE, it returns TRUE else FALSE.

Syntax:

=OR(*Cond1,Cond2*) returns *TRUE or FALSE*

"SUM": It is a function under the math category. It returns the sum total of all the values passed to it.

Syntax:

=SUM(*Num1,Num2...*) returns *Num1+Num2+....* **OR**

=SUM(*Cell Range*) returns *Sum of all the cells in Cell Range*

"AVERAGE": It is a function under the statistical category. It returns the arithmetic mean of the numbers provided. That is, it adds all the numbers and divides the total by the count of numbers.

Syntax:

=AVERAGE(*Num1, Num2, ...*) returns (*Num1 + Num2 + ...*)/Count

OR

=AVERAGE(*Cell Range*) returns the average of all the numeric values in the specified range.

"COUNT": It is a function under the math category. It returns the count of the values given as parameters.

Syntax:

=COUNT(*Num1,Num2...*) returns *Count of Num1, Num2,....*

=COUNT(*Cell Range*) returns *Count of all the cells in Cell Range*

"IF": It is a function under the Logical category. It takes logical condition/s and their respective outcomes. Based on the evaluation of the logical condition/s, the output is produced.

Syntax:

=IF(*Cond,value_if_true,value_if_False*) returns *value_if_true or value_if_False*

"SUMIF": It is a function under the math category.

Syntax:

=SUMIF(*Cell Range, condition*) returns *Sum of all the cell in Cell Range which meets the given criteria*

"COUNTIF": It is a function under the math category.

Syntax:

=COUNTIF(*Cell Range, Criteria*) returns *Count of all the cells in Cell Range which meets the given criteria*

"INDEX": It is a function under the Lookup and reference category.

Syntax:

=INDEX(*Cell*) returns *Content in the given cell*

=INDEX(*Cell Range,row,column*) returns *Content in the cell(row,col)*

List of Useful Operators

- Sum: +
- Subtraction: -
- Multiplication: *
- Division: /

- Exponential: ^
- Fix a column or row: \$
- Equal sign: =
- Greater-than sign: >
- Lesser-than sign: <
- Greater-than-or-equal sign: >=
- Lesser-than-or-equal sign: <=
- Not-equal to sign: <>

Table of Excel Shortcuts

Shortcut	Action	Category
Ctrl + C	Copy	Ctrl
Ctrl + V	Paste	Ctrl
Ctrl + X	Cut	Ctrl
Ctrl + Z	Undo	Ctrl
Ctrl + Y	Redo	Ctrl
Ctrl + A	Select all	Ctrl
Ctrl + S	Save	Ctrl
Ctrl + F	Find	Ctrl
Ctrl + H	Replace	Ctrl
Ctrl + Arrow Key	Jump to edge of data region	Ctrl
Shift + Arrow Key	Extend selection one cell at a time	Shift
Shift + Space	Select entire row	Shift
Ctrl + Space	Select entire column	Ctrl + Shift
Shift + F11	Insert new worksheet	Shift
Shift + Enter	Enter value and move up	Shift
Ctrl + Shift + L	Toggle filter on/off	Ctrl + Shift
Ctrl + Shift + Arrow	Select entire block of data	Ctrl + Shift
Ctrl + Shift + "+"	Insert new row or column	Ctrl + Shift
Ctrl + Shift + "\$"	Apply currency format	Ctrl + Shift
Ctrl + Shift + "%"	Apply percentage format	Ctrl + Shift
Alt + =	AutoSum	Other
F2	Edit active cell	Other
Ctrl + Tab	Switch between open workbooks	Other
Ctrl + Page Up/Down	Move between worksheets	Other